

Heteroclinics and Clean Periodic Rotations of the Suslov Body

Route-A source-local certificate HCS-C255

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Abstract

We give a complete positive-energy reduced atlas for the classical Suslov rigid body. Every generic energy ellipse has two explicit heteroclinics and two permanent rotations; the latter reconstruct as clean periodic $SO(3)$ families with an exact primitive attitude period. This source-local nonholonomic theorem has no arithmetic origin.

1 Constraint and reduced equations

Align the forbidden body axis with e_3 and rotate the allowed plane so that

$$I = \begin{pmatrix} I_1 & 0 & a \\ 0 & I_2 & b \\ a & b & I_3 \end{pmatrix}, \quad I_1, I_2 > 0, \quad I_3 > \frac{a^2}{I_1} + \frac{b^2}{I_2}. \quad (1)$$

The constraint is $\omega_3 = 0$, and $I\dot{\omega} = (I\omega) \times \omega + \lambda e_3$. With $\ell = a\omega_1 + b\omega_2$, the first two components are

$$I_1\dot{\omega}_1 = -\ell\omega_2, \quad I_2\dot{\omega}_2 = \ell\omega_1, \quad H = \frac{1}{2}(I_1\omega_1^2 + I_2\omega_2^2). \quad (2)$$

Direct substitution gives $\dot{H} = 0$.

2 Both heteroclinic components

Set $u = \sqrt{I_1}\omega_1$, $v = \sqrt{I_2}\omega_2$ and $(u, v) = R(\cos \phi, \sin \phi)$, $R = \sqrt{2H}$. For

$$q^2 = \frac{a^2}{I_1} + \frac{b^2}{I_2}, \quad \cos \delta = \frac{a}{\sqrt{I_1}q}, \quad \sin \delta = \frac{b}{\sqrt{I_2}q}, \quad \kappa = \frac{Rq}{\sqrt{I_1I_2}}, \quad (3)$$

equations (2) reduce to $\dot{\phi} = \kappa \cos(\phi - \delta)$.

Theorem 1. *If $q, H > 0$, the two components complementary to $\ell = 0$ are*

$$\sin(\phi - \delta) = \tanh \kappa(t - t_0), \quad \cos(\phi - \delta) = \sigma \operatorname{sech} \kappa(t - t_0), \quad \sigma = \pm 1. \quad (4)$$

Both run from the reduced-unstable permanent rotation to its antipodal reduced-stable partner, with exponents $+\kappa$ and $-\kappa$.

The identities $\tanh^2 z + \operatorname{sech}^2 z = 1$ and differentiation of (4) prove the theorem. The scaled angular deflection is π .

3 Permanent rotations and boundaries

At $\ell = 0$, angular velocity is constant. The full attitude therefore is

$$Q(t) = Q_0 \exp(t\hat{\omega}), \quad T = \frac{2\pi}{|\omega|} \quad (\omega \neq 0). \quad (5)$$

More explicitly, for $q > 0$ the unstable/stable endpoints are

$$\omega_{u,s} = \pm \sqrt{\frac{2H}{I_1b^2 + I_2a^2}} (b, -a, 0), \quad T = \frac{2\pi\sqrt{I_1b^2 + I_2a^2}}{\sqrt{2H}\sqrt{a^2 + b^2}}. \quad (6)$$

Thus every generic energy ellipse already has periodic full-flow rotations; they must not be erased by the heteroclinic statement. For each endpoint, the distinct attitude circles are the right cosets $SO(3)/SO(2) \simeq S^2$, hence a clean two-dimensional family; energy adds a continuum. These analytic periodic families justify **A1_WEAK**, not an isolated-orbit pass. If $a = b = 0$, every reduced state is permanent and every nonzero full state is periodic. If only one of a, b vanishes, (3)–(4) remain generic. At $H = 0$ the attitude is fixed.

4 Reduced Poisson structure and reversor

The two-dimensional reduced flow is generated by H through

$$\{\omega_1, \omega_2\} = -\frac{\ell}{I_1 I_2}. \quad (7)$$

Every bivector in dimension two satisfies Jacobi. On either open half-plane $\ell \neq 0$, division by $|\ell|$ leaves a divergence-free linear rotation, so $d\omega_1 d\omega_2 / |\ell|$ is invariant. At a nonzero permanent rotation the original divergence is nonzero; consequently no positive C^1 invariant density can extend through that endpoint. The full involution $(Q, \omega) \mapsto (Q, -\omega)$ reverses time because the reduced quadratic vector field is even.

face	reduced behavior	full reconstruction
$q > 0, \ell \neq 0$	two heteroclinic arcs	nonrecurrent state
$q > 0, \ell = 0$	two permanent endpoints	periodic rotations
$q = 0, H > 0$	entire ellipse permanent	periodic rotations
$H = 0$	origin	fixed attitude

5 Executable boundary

The exact certificate stores twelve generic and four principal-axis inertia rows. Rational squares record the radical scales. An independent checker closes 244 assertions; fresh SymPy closes 18 identities; replay is byte-exact; and 30/30 rehashed semantic mutations are rejected, including denial of the generic clean periodic family. The 2024 geometric source owns the reduced equations and bracket, while the 2009 and 1995 sources supply reconstruction and rigid-body context. No literature-priority claim is made.

6 Route-A stop

The arithmetic origin is **none**. The tuple is (**A0_FAIL**, **A1_WEAK**, **A2_FAIL**, **A3_FAIL**, **A4_FORMAL_HINT**) and the verdict is **ROUTE_A_REJECTED**, with **route_b_invocation_allowed**: **false**, under **NO_BAD_EULER_OR_ROOT_NUMBER**. No target determinant or Hilbert–Pólya operator is claimed. The bracket (7) degenerates on the periodic set and is only an **A4_FORMAL_HINT**; no auxiliary Hamiltonized clock is substituted for physical time.

References

- [1] L. C. García-Naranjo, J. C. Marrero, D. Martín de Diego and P. E. Petit Valdés, *J. Nonlinear Sci.* 34 (2024), article 110.
- [2] Y. N. Fedorov, A. J. Maciejewski and M. Przybylska, *Nonlinearity* 22 (2009), 2231–2259.
- [3] Y. N. Fedorov and V. V. Kozlov, *AMS Transl.* 168 (1995), 141–171.